Project 3: Ion Drive Rocket Analysis

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The purpose of this project is to

Nomenclature

T	=	temperature $[K]$
P	=	pressure $[Pa]$
ρ	=	density $\left[kg/m^3\right]$
T_t	=	total temperature $[K]$
P_t	=	total pressure $[Pa]$
u	=	velocity $[m/s]$
M	=	mach number
A	=	area $\left[m^2\right]$
F	=	thrust $[N]$
R	=	gas constant $[J/kgK]$
C_p	=	constant pressure specific heat $[J/kgK]$
γ	=	specific heat ratio
$\dot{m}$	=	mass flow rate $[kg/s]$
I_{sp}	=	specific impulse $[s]$
g_0	=	standard gravitational acceleration $\left[9.81m/s^2\right]$

I. Introduction

The purpose of this project temperature and pressure information is shown in table **??**. This data was used when running the solver at each altitude.

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II. Rocket Analysis

The following equation can be used to calculate the propellant mass required to perform a maneuver.

$$\frac{\Delta v}{C} = \ln\left(\frac{m_0}{m_0 - m_p}\right) - \int_{t_0}^{t_f} \frac{D}{Cm} dt - \int_{t_0}^{t_f} \frac{g\cos\theta}{C} dt \tag{1}$$

Because the drag term and the gravity can be ignored, this equation can be simplified to the following.

$$m_p = m_0\left(1 - e^{\frac{-\Delta v}{C}}\right) \tag{2}$$

C in this equation is the effective exhaust velocity which can be found by using the following.

$$C = I_{sp} g_0 = 34335.0\ m/s \tag{3}$$

The required propellant mass for each stage of the flight is:

$$m_{p_a} = \boxed{51,134.167\ kg}$$

$$m_{p_d} = \boxed{18,450.379\ kg}$$

This gives a total propellant mass of:

$$m_p = 69584.546\ kg$$

Using the assumption that after the deceleration all the propellant is spent (the only remaining mass is that of the engine and payload), knowledge of the initial mass and payload mass can be used to calculate the engine mass. The engine mass m_e can be calculated using the equation below. Here m_0 is the initial mass, m_p is the propellant mass, and m_L is the payload mass.

$$m_e = m_0 - (m_p + m_L) = \boxed{5415.454\ kg} \tag{4}$$

This engine mass is then used in subsequent ion drive calculations.

III. Ion Drive Analysis

Using the engine mass calculated previously the total engine power, accelerator and ionizer, can be calculated given that the engine power per unit engine mass is 600 Watts per kilogram.

$$W_{engine} = \boxed{3,249,272\ W}$$

Since it is known that the ionizer uses 35% of the power that the accelerator uses, the accelerator power can be found by using the following:

$$W_{accelerator} = W_{ion} = \frac{W_{engine}}{1 + 35/100} = 2,406,868\ W \tag{5}$$

The effective exhaust velocity in the specific impulse equation can be rewritten in terms of propellant charge per mass and potential difference. Solving for the potential difference gives the voltage required by the accelerator.

$$I_{sp} = \frac{1}{g_0}\sqrt{2\frac{q}{m}(\phi_A - \phi_C)} \rightarrow (\phi_A - \phi_C) = V = \boxed{235,778\ V} \tag{6}$$

Using the accelerator power and voltage, the required current can be calculated:

$$I = \frac{W_{ion}}{V} = \boxed{10.208\ A} \tag{7}$$

Using the maximum allowable voltage gradient the distance between the anode and cathode can be found:

$$\frac{(\phi_A - \phi_C)}{d} = 10^5\ \frac{V}{cm} \rightarrow d = \boxed{2.35778\ cm} \tag{8}$$

To find the beam area the equation for current per cross sectional area can be used. Because the assumption that the maximum possible current is being used the following can be done:

$$j_{max} = 0.4444\epsilon_0\sqrt{2\frac{q}{m}}\frac{(\phi_A - \phi_C)^{3/2}}{d^2} = 57.273\ \frac{A}{m^2} \tag{9}$$

Where ϵ_0 is the universal constant of permittivity of free space $8.85 \times 10^{-12}\ \frac{C^2}{Nm^2}$

The area is then:

$$A_{beam} = \frac{I}{j_{max}} = \boxed{1782.37\ cm^2 = 0.178237\ m^2} \tag{10}$$

IV. Brayton Cycle Analysis

$$\pi_t = \frac{1}{\pi_c} \tag{11}$$

$$\frac{T_4}{T_3} = 1 - \eta_t \left(1 - \pi_t^{\frac{\gamma-1}{\gamma}}\right) \tag{12}$$

$$\frac{T_2}{T_a} = 1 + \frac{1}{\eta_c} \left(\pi_c^{\frac{\gamma-1}{\gamma}} - 1\right) \tag{13}$$

$$\eta = 1 - \frac{T_4 - T_a}{T_3 - T_2} \tag{14}$$

$$\frac{A_{rad}}{W_{out}} = \frac{\left(\frac{1}{\eta} - 1\right)}{3(T_4 - T_a)\sigma\epsilon} \left(\frac{1}{T_a^3} - \frac{1}{T_4^3}\right) \tag{15}$$

V. Results

VI. Appendix A